

Township of
NIPISSING

"Life the way it should be" ESTABLISHED 1888

STAFF REPORT

LANDFILL STATUS 2023



Landfill Status Report 2023

Background:

The 2019 Landfill Annual Monitoring Reports were received from Aecom and revealed some concerning information with special regard to the Bear Creek Landfill.

Aecom was then asked for Landfill Footprint Staking and Cell Sequencing/Fill Plan Works for both landfill sites in April 2020. With this information it was determined that the Bear Creek Landfill had fill placed outside the allowable limits. Corrective action was taken.

With this information, the needs for increased vigilance on the landfill operations, monitoring and waste disposal/diversion activities were identified.

Remaining Life Calculations

Bear Creek Landfill

Year	m^3 Deposited	m^3 Airspace Remaining	m^3 Average Deposited	Years Remaining
2010	696	18,589		26.7
2017	790	17,744	502*	35
2018	150	17,594	382**	45
2019	585	17,009	457***	37

A review in 2014 identified additional 5,300 m^3 fill area, increasing remaining Airspace to 19,750 m^3.
* 2017 Averaged over 4 years = 2014, 2015, 2016, 2017
** 2018 Averaged over 5 years = 2014, 2015, 2016, 2017, 2018
*** 2019 Averaged over 6 years = 2014, 2015, 2016, 2017, 2018, 2019

Figures prior to 2014 were calculated using a different method. Calculations are now made using a volume survey.

Note: Staff reported concerns about the estimated remaining years stated for the Bear Creek Landfill. The fill rate is quite high at this site and the remaining fill space did not appear to be as large as has been used as an operating guide in previous years.

Recommendations:

1. Increase the reserve amounts going toward landfill closure to the $20,000 per year in order to ensure there are adequate funds for closure sooner than anticipated.

2. Divert all but household waste and commercial waste from this location.

3. Begin to limit the amount of household waste permitted by residence in order to reduce the amount entering the site.

Wolfe Lake Landfill

Year	m^3 Deposited	m^3 Airspace Remaining	m^3 Average Deposited	Years Remaining
2010	1,395	30,133		24
2017	387	44,100	400*	110
2018	220	43,880	362**	120
2019	575	43,305	399***	108

* 2017 Averaged over 4 years = 2014, 2015, 2016, 2017
** 2018 Averaged over 5 years = 2014, 2015, 2016, 2017, 2018
*** 2019 Averaged over 6 years = 2014, 2015, 2016, 2017, 2018, 2019
Figures prior to 2014 were calculated using a different method. Calculations are now made using a volume survey.

Options:

1. Place a 2 bag limit on each household per week and implement a tracking system for this method. The punch card system is a basic way of monitoring this.
   Concerns have been expressed regarding tourist operations and a small business collecting household waste within the Township and how to manage a limitation on waste amounts. Staff can work with the small business owner collecting waste to ensure that this system can be implemented and maintain their business viability. Limit variations can also be implemented for tourist operators to ease the transition.

2. Limit the number of vehicles accessing the site. In order for staff to effectively inspect the waste entering the landfill sites, it is important to limit the number of vehicles at the site at any time. Having 2 staff members, 1 can work from the landfill building greeting vehicles and confirming Landfill Entry Permit is available. Sending vehicles to the active fill area to be met by a staff member who can then inspect the load coming in, educate and direct people to the proper disposal areas and enforce the rules of the landfill site.

2023 UPDATE

Following an RFP process for Landfill Monitoring Services, Council awarded Pinchin Consulting the contract effective for the 2022 monitoring year.

2022 monitoring revealed

Remaining Life Calculations

Bear Creek Landfill

Year	m^3 Deposited	m^3 Airspace Remaining	Years Remaining
2022	2,850	12,422	4.36

Wolfe Lake Landfill

Year	m^3 Deposited	m^3 Airspace Remaining	Years Remaining
2022	2,400	41,990	17.5

Traffic was rerouted and greeted by a Staff member upon entry. Landfill Entry permits have been numbered and are recorded with bag counts upon each visit.

Bag limits have not been imposed.

Landfill Operations

According to the Landfill reports, compaction was occurring each day of operation at each landfill using the Township Excavator moving over the waste at the end of the day for 2017.

In 2018 and 2019 it was reported to be happening once per week at each Landfill.

For proper compaction of the waste at each landfill site, compaction equipment upgrades are required. The Excavator is not sufficient in weight to accomplish what is required to give the most compaction and obtain the optimal airspace at the sites.

Evan Hughes Excavating performed compaction services at both landfill sites in 2017, 2018 and 2019.

2017	$ 4,202.69 (tax rebate removed)	Compaction service 1 time.
2018	$11,282.01 (tax rebate removed)	Compaction services 2 times.
2019	$ 6,466.84 (tax rebate removed)	Compaction services 2 times.

**Costs also include cover material costs, obtained as required.*

Recommendation:

Each Landfill Site requires complete compaction monthly to maintain optimum compaction and airspace rates.

Options:

1. Estimated costs for compaction services monthly, Evan Hughes Excavating: $2,500.00, resulting in approximately $30,000 per year.

2. Purchase a refurbished compactor unit and new float unit for transport between sites.

   Estimated cost of used equipment: $250,000.00
   Estimated annual service costs: $10,000.00
   This unit and float could be financed over a five-year period or use of the modernization funding can be applied.

Staff Recommends moving forward with using local services, #1, with the benefits of not having to maintain the equipment, train staff, maintain liability for the work and by using local services we are contributing to the local economy for a specialized service.

2023 UPDATE

A compactor was purchased in early 2023 for use at each landfill site. The results of the operation of compaction will be available with the remaining airspace calculations provided by Pinchin for 2023.

Grinding Services

The Township brings in Sittler Grinding Services to grind up the large materials such as mattresses, furniture and large construction debris. The shredded material is then used for landfill cover. Having all large items brought only to Wolfe Lake Landfill reduces staff time moving large items between landfill sites and removes all costs associated with having the grinding machine attend each site.

Year	Cost of Grinding Services
2018	$13,330.56
2019	$16,993.92

We have reached out to the City of North Bay for information on the mattress recycling program they have in place. Working with a company from Montreal, the mattresses collected for the City of North Bay are then diverted from the landfill site and used for other purposes.

A request for costing and material storage requirements has been sent and we are waiting on a response. Due to the Coronavirus Emergency, not all departments are able to respond to us at this time.

2023 UPDATE

Year	Cost of Grinding Services
2020	$11,702.40
2021	$17,299.20
2022	$25,918.27
2023	$20,784.00

Staff have been researching and requesting quotations for service from alternate service operators as they become available.

Mattress Recycling

In 2022 mattress diversion was started with a working agreement from a company located in Montreal, Quebec. The cost for the service in 2022 was $6,921.77 with approximately 275 mattresses diverted from the landfill and removed from the shredding program. Current User Fees of $25 per mattress/box spring, cover the cost of this service for diversion. Mattresses have been picked up for the 2023 program. 114 mattresses, total cost of pick up $2,868 before taxes. This averages to $25.16 per mattress for diversion which remains in line with the user fee being charged for disposal.

Recycling Services

Blue Box Materials are collected by R&D Recycling using bins located at each landfill site. When a pick up is required, the Township contacts R&D Recycling to come out.

The material collected is not weighed at a station and weigh tickets are not supplied. The total tons collected are provided at month end and at year end a product breakdown is provided by R&D Recycling for reporting purposes. The Township does not operate under a contract for this service.

With the Waste Diversion and Circular Economy Act, the costs associated with recycling are to be covered by the producer of the product. This is still being organized to see how this will be obtained. Township Staff continues to work with RPRA in order to ensure we are included in the program changes.

Costs for Recycling Services:

Year	Paid for Services (less tax rebate)
2017	$54,014.06
2018	$59,673.89
2019	$50,630.57

2023 UPDATE

An RFP for Recycling Services was circulated in 2021, effective January 1, 2022. Canor was the successful bidder for Recycling services.

Year	Paid for Services (less tax rebate)
2020	$47,940.87
2021	$46,315.35
2022	$64,892.32

Waste Diversion Figures

Stewardship Ontario provides an annual amount to assist with waste diversion costs at the landfill sites. This funding is collected from the producers of recyclable materials.

Year	Amount Received from Stewardship Ont.
2016	$33,767.40
2017	$36,924.40
2018	$34,577.04
2019	$25,167.72

Each year we submit all diverted waste amounts to Stewardship Ontario and they advise of the portion of payment we will receive. Markets have changed for the recycling industry and there is not as much revenue available as in previous years. The Ontario Government has put in place the Waste Diversion and Circular Economy Act

2023 UPDATE

Year	Amount Received from Stewardship Ont.
2020	$27,676.20
2021	$26,549.42
2022	$25,431.09
2023	$30,598.52

The Township also receives funding from Ontario Electronic Stewardship (OES) for the electronic waste collected at each site and diverted from the landfill sites.

Year	Amount Received from OES
2017	$ 976.50
2018	$1,824.00
2019	$2,113.50

OES will cease to exist as of December 31, 2020 and an agreement is in place for the Electronic Products Recycling Association (EPRA) to continue this service.

2023 UPDATE

Year	Amount Received from EPRA
2020	$1,905.00
2021	$1,578.75
2022	$1,090.50

In 2019 the Township began taking the steel from the landfill to the scrap dealer and keeping all profit.

Year	Amount Received from Scrap Metal
2019	$ 762.80
2020 up to May	$2,592.61

Appliances containing freon, such as refrigerators and freezers, are accepted at each site. If they are not tagged as freon free by a registered operator, the cost is $35.00 per unit. Twice per year an authorized operator is brought in to remove the freon from these units, providing the Township with the required records of safe, authorized removal, and then the remaining appliances are taken with scrap loads.

The company we use currently will no longer be in business for future needs so we anticipate a slight cost increase as we obtain a new operator.

The cost of having the registered operator service come to the sites has averaged $25.00 per unit with the remainder of the service fee towards the costs of moving the units to the scrap dealer with our fleet vehicles and staff.

2023 UPDATE

Year	Amount Received from Scrap Metal
2020	$11,037.28
2021	$14,766.35
2022	$16,962.13
2023 up to Aug 31	$12,297.93

Tipping Fees collected:

Year	Tipping Fees
2017	$9,869.50
2018	$13,158.00
2019	$14,698.00

User Fees Current:

LANDFILL	
Household Appliances	$ 10.00
Fridge, freezer or air conditioner – tagged as Freon free	$ 10.00
- untagged	$ 40.00
Brush – 5 ton dump trailer and under	$ 5.00
BULK items accepted at Wolfe Lake Landfill ONLY	
Mattress or box spring	$ 25.00
Carpet, flooring up to 12' x 12'	$ 25.00
Carpet, flooring over 12' x 12'	$ 50.00
Couch	$ 20.00
Stuffed or upholstered chair/ottoman	$ 10.00
Plastic Outdoor Furniture (per item)	$ 5.00
Tarp (10' x 20' and over)	$ 5.00
Shingles – 5-ton dump trailer and under	$ 30.00
Renovation waste – 5-ton dump trailer and under	$ 75.00
Watercraft	$ 150.00
MIXED LOADS MUST BE SORTED FOR DISPOSAL:	
Tandem Trailer	$ 80.00
Single Axle (Dump Truck)	$ 180.00
Tandem Truckloads	$ 250.00
Dumpster or Tri-Axle Truckloads (12 Yards and Up)	$ 300.00

User Fees in 2019 with recommendations:

LANDFILL	Proposed	2019
Household Appliances	$ 8.00	$ 5.00
Fridge, freezer or air conditioner – tagged as Freon free	$ 8.00	$ 5.00
- untagged	$ 38.00	$ 35.00
Brush – 5 ton dump trailer and under	$ 10.00	$ 5.00
BULK items accepted at Wolfe Lake Landfill ONLY		
Mattress or box spring	$ 25.00	$ 5.00
Shingles – 5-ton dump trailer and under	$ 30.00	$ 25.00
Renovation waste – 5-ton dump trailer and under	$ 75.00	$ 75.00
Single Axle	$ 180.00	$110.00
Tandem Truckloads	$ 250.00	$180.00
Dumpster or Tri-Axle Truckloads (12 Yards and Up)	$ 250.00	$180.00
Tandem Trailer	$ 80.00	$ 50.00
Watercraft	$ 150.00	$150.00

2023 UPDATE

Tipping Fees collected:

Year	Tipping Fees
2020	$20,902.50
2021	$28,539.62
2022	$27,439.65

Textile Bins

Through an agreement with the Diabetes Canada, bins are maintained at each landfill site to collect textiles. A total tonnage amount is provided to us each year for reporting to Waste Diversion Ontario to be included in our statistics.

In 2019 4,458 pounds of textiles were diverted from the landfill sites.

**Due to the 2020 Coronovirus Emergency Declaration, the service trucks used for the textile diversion program were deemed non-essential and have been unable to pick up textiles since the March 2020 Emergency Declaration. This will reduce figures for 2020.*

2023 UPDATE

Year	Textiles Diverted (converted to Tonnes)
2018	2.76
2019	2.02
2020	.93
2021	.31
2022	2.91
2023 up to Aug 31	1.49

FINANCIAL BREAKDOWN COMPARISON

Revenue			
Year	**Tipping Fees**	**Diversion**	**Total**
2017	$9,869.50	$37,111.65	$46,981.15
2018	$13,158.00	$36,987.88	$49,559.04
2019	$14,698.00	$30,393.35	$42,742.02
2020	$20,902.50	$40,618.48	$61,520.98
2021	$28,539.62	$44,892.52	$73,432.14
2022	$27,439.65	$43,483.72	$70,923.37

Expenses				
Year	**Landfill***	**Recycling**	**Monitoring**	**Total**
2017	$7,866.76	$54,014.06	$50,300.00	$112,180.82
2018	$30,795.99	$59,673.89	$35,740.35	$126,210.23
2019	$30,946.65	$50,630.57	$43,912.00	$125,489.22
2020	$38,117.57	$47,940.87	$58,410.24	$144,468.68
2021	$34,821.72	$46,315.35	$41,263.68	$122,400.75
2022	$48,135.05	$64,892.32	$30,851.60	$143,878.97

Landfill* Expenses are for materials and operations only. Does not include any staff or equipment expenses.

Costs associated with capital improvements and modernization grant expenses are not included in these figures.

Tipping Fees for items that require grinding services, qualified operator services to de-gas and the use of staff and Fire Department services, generate revenue to help offset costs associated with those activities over and above the management of household waste and recycling.

Public Education Tools and Methods Used

Township Newsletters are utilized to distribute education material and encourage participating in diversion programs. The Ad Hoc Landfill Committee also created and circulated a dedicated information newsletter for this purpose in 2020.

A Town Hall meeting was held in 2012 to announce the Clear Bag initiative. A Survey was also circulated to all residents on paper and online to allow for feedback and provide a list of educational information we would require to have circulated.

A Town Hall meeting was also held in June 2019 to discuss waste diversion and landfill operations, among other topics. Owner of R&D Recycling was present to provide a detailed description of recycling services. This meeting was not well attended.

Recommendations:

Continue to education residents using the Newsletters and Public Meetings.

2023 Recommendations:

1. **Impose bag limits per year based on property use and assessment status.**
2. **Move to a 4 day work week schedule for Landfill locations with each being open 2 days per week, 10 hour days. The operating hours of the landfill locations remains the same with a different schedule to allow for change in usage.**
3. **Investigate options for diverting more bulk items as the industries open up.**
4. **Initiate an education program for residents on composting methods and investigate ways to assist residents compost organic material and divert it from the landfill sites.**